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Cloud39EV300 Platform Development Board Instructions

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Contact

Ankai (Guangzhou) Microelectronics Technology Co., Ltd.

Address: 3rd Floor, C1 Zone, Innovation Building, No. 182, Science Avenue, Guangzhou Science City

Tel: (86)-20-3221 9000

Fax: (86)-20-3221 9258

Postal Code: 510663

Sales Hotline:

(86)-20-3221 9499

Email:

sales@anyka.com

Home:

<http://www.anyka.com>

Version Change Description

The following table briefly describes the version changes of this document. Version changes are limited to changes in technical content and do not include changes in layout, format, syntax, etc.

Version	Illustrate	Completion Date
V1.0.0	Official Release	December 2017
V1.1.0	Modify key device support	May 2018

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1 Document Introduction

This document mainly introduces the interface, burning process, video preview, information configuration and other functions of the development board in detail to help R&D personnel quickly become familiar with the operation process of the development board.

2 How to quickly generate video on the development board

The five simple steps to produce a video on the development board are as follows.

(1) Check the board: First, make sure that the board you have is complete. At least make sure that there are a baseboard, a core board, a PHY board, and a sensor board. The actual picture is shown below.

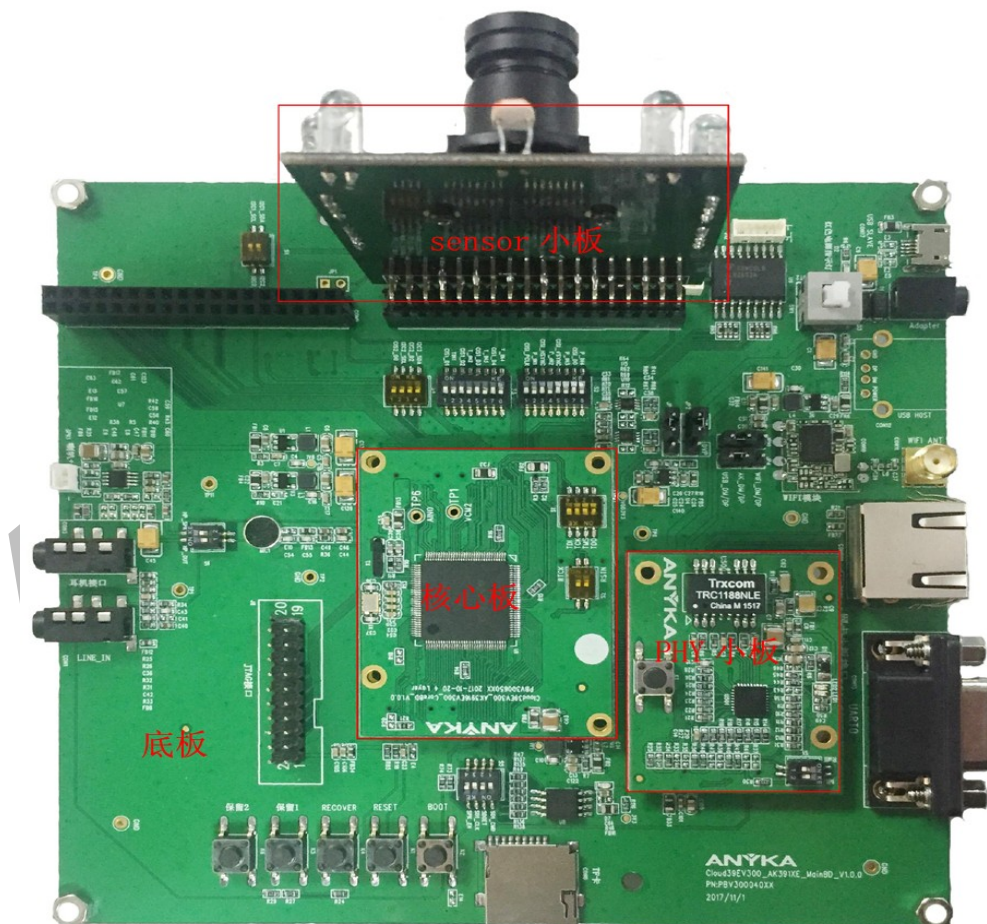


Figure 2-1 Complete development board

(1) Find the address information of the development board: Connect the development board to a 5V USB for power supply, connect the development board to the PC via a network

cable, and power on the device. Run the ONVIF Device Manager tool on the PC, as shown in the figure below. The device list is on the far left. Select the device list, then click the Live video option, and you can see the development board address "rtsp://192.168.1.88:554/ch0_0.264" at the bottom left of the page.

Note: Please make sure that the IP address of the PC is in the same network segment as the IP address of the development board. Since ONVIF Device Manager does not support H.265 video decoding, it does not support video preview and is only used to find address information.

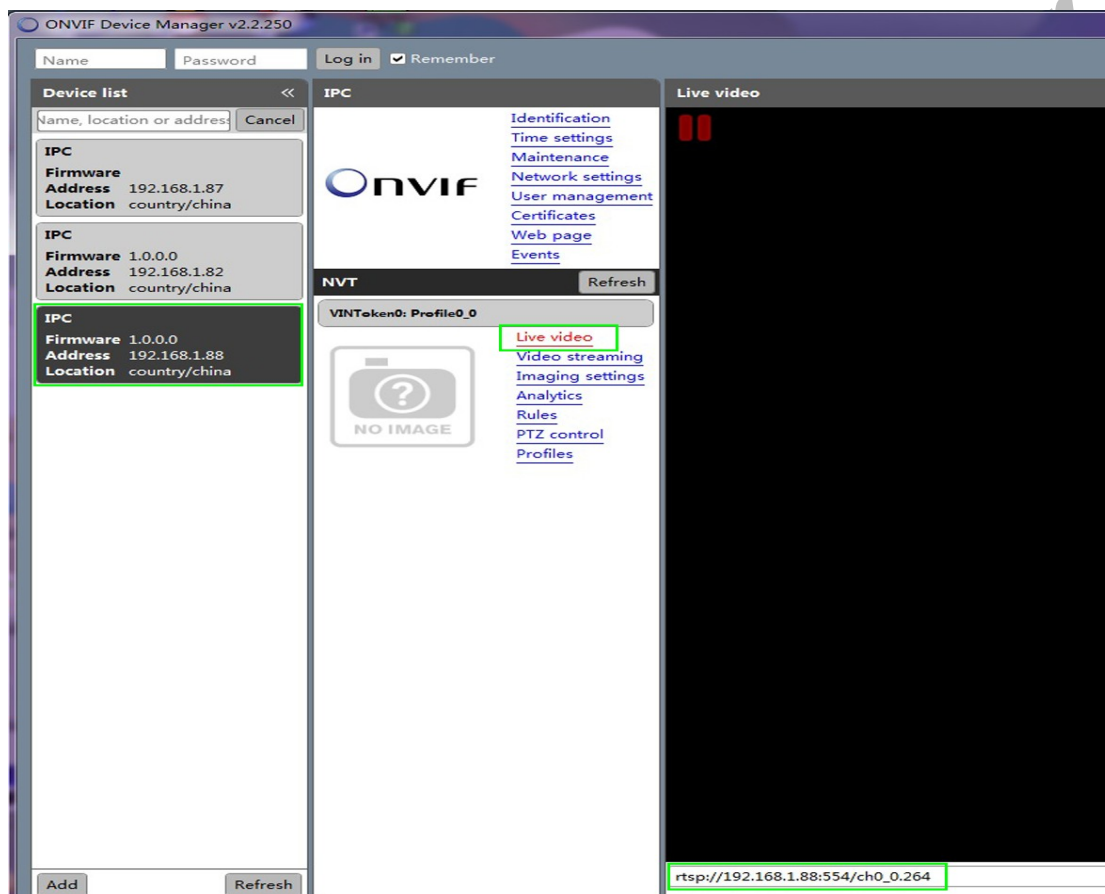


Figure 2-2 ONVIF Device Manager search image

(2) (2) Install VLC or Potplayer software on the PC: VLC media player V 2.2.6 or potplayer V1.7.4597 or above, as shown in the figure below (both software can be downloaded directly from the Internet).

(3) Note: Please confirm the version number of VLC and Potplayer, otherwise unknown errors may occur.

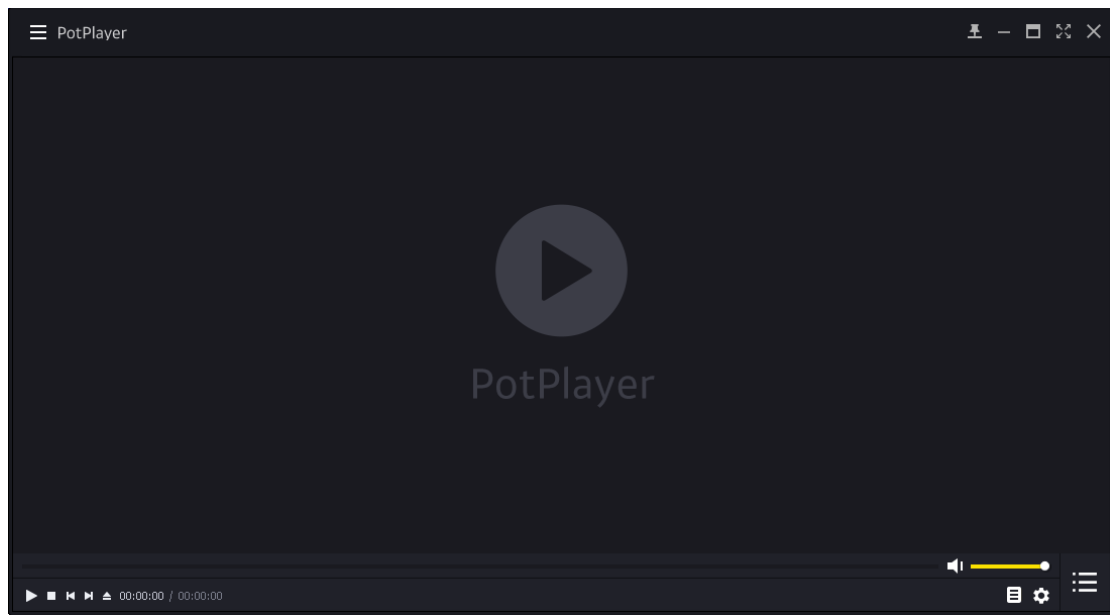


Figure 2-3 Potplayer interface

(4) Configure the software (VLC and Potplayer are configured in the same way, only Potplayer is used as an example): Open Potplayer, select the "Open" option in the main menu bar, then select "Open Link", enter the address information "rtsp://admin:@192.168.1.88:554/ch0_0.264" obtained in step (1) in the interface, and click the OK button to complete the configuration. Please note that you should add "admin:@" before the address. The operation interface is shown in the figure below.



Figure 2-4 Open link

(5) Preview completed: At this point, the quick preview operation is completed, and the video preview is shown in the figure below.

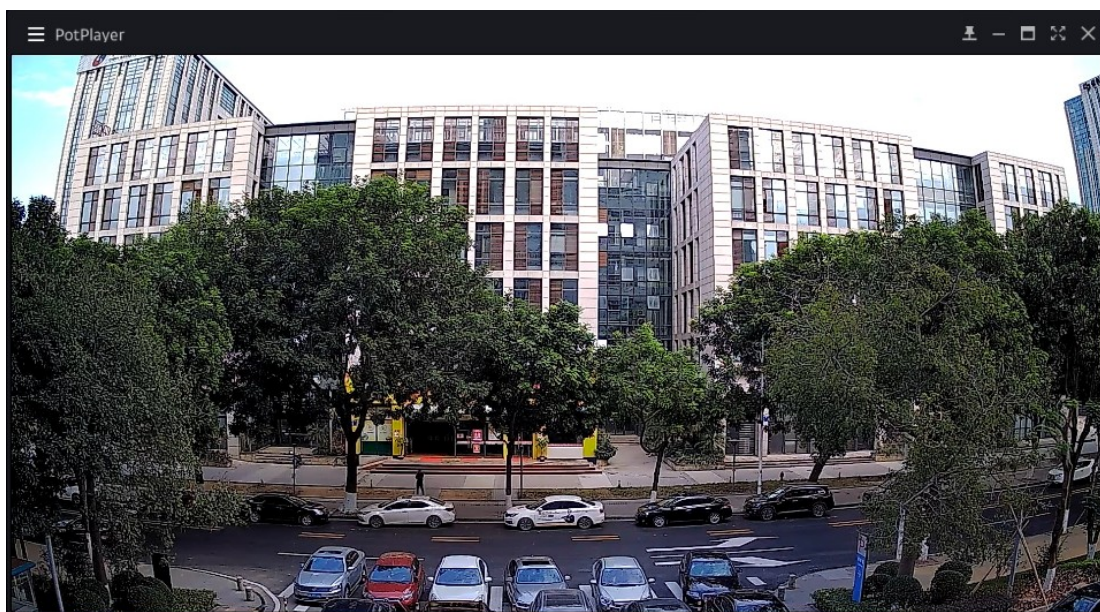


Figure 2-5 Potplayer tool video preview

3 Development Board Introduction

The development board of the Cloud39EV300 platform is mainly composed of a baseboard, a core board, a sensor board, a PHY board, a Wi-Fi board and other key components. The baseboard provides USB Wi-Fi, various peripheral interfaces and function keys; the core board integrates the main control chip and crystal oscillator; the peripheral boards are mainly for sensors, Wi-Fi modules and PHY chips.

3.1 Development board components

The components of the development board are shown in the following table.

Table 3-1 Development board components

Number	Part	Specification	Quantity
1	Base Plate	Platform base	1
2	Core board	AK3916EV300 Main control chip	1
		AK3918EV300 Main control chip	
3	Sensor board	SC2235	1
		SC2232	
		IMX323	
		GC2033	

4	PHY board	KSZ8041NLI	1
		IP101GRI	
		RTL8201F	
6	USB Wi-Fi Module	8188EUS	1
		8188FTV	
7	SPI NOR Flash	GD25Q64C	1
		GD25Q127C	
8	Fixing posts and nuts	Fix the development board base	4

3.2 Development board interface and functions

The physical diagram of the complete development board is shown below.

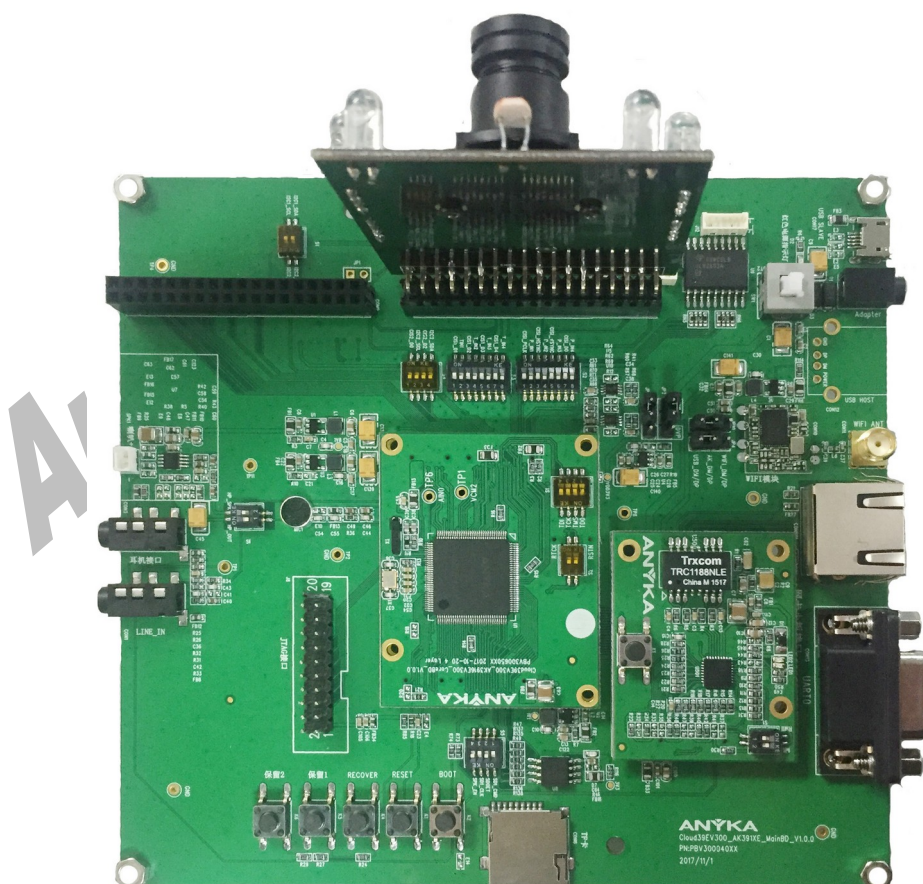


Figure 3-1 Physical picture of the development board

3.2.1 Development board base

The baseboard of the development board mainly provides external interfaces and function buttons. The actual picture is shown in the figure below.

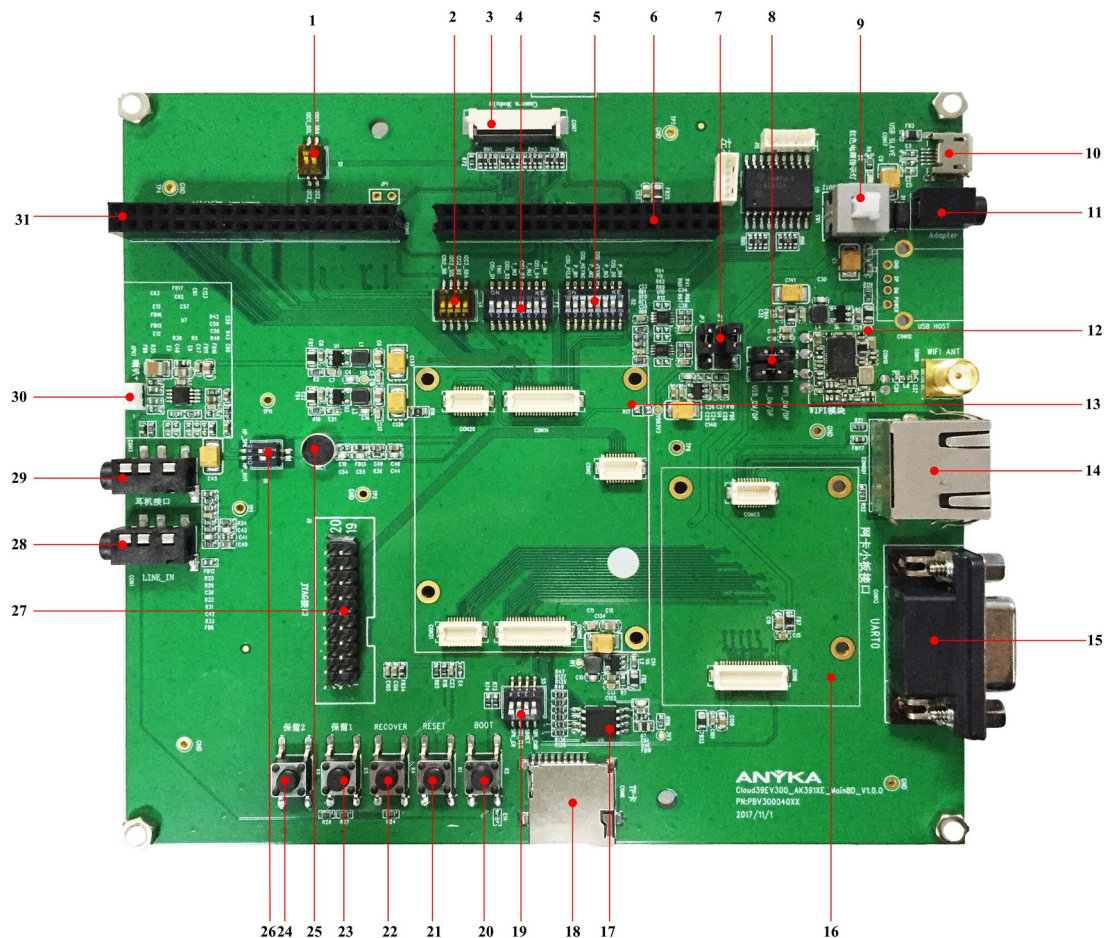


Figure 3-2 Development board base

3.2.1.1 The interfaces and functions of the baseboard are described in the following table.

Table 3-2 Bottom panel ports and functions

Number	Name	Illustrate
1	S1 switch	Camera2's TWI interface selection switch selects one of TWI1 and TWI2 as the interface for initializing the camera2 sensor.
2	S4 switch	Function selection switch of GPIO52 and GPIO53 interfaces, select as data line D0 and D2 of camera2 sensor, or as TWI2 control interface. Note: S1 and S4 need to be used together.
3	CON7 Sensor	Aerial sensor interface.

Number	Name	Illustrate
	interface	
4	S3 switch	Function selection switch of GPIO7, GPIO8, GPIO9, and GPIO68 interfaces, which can be selected as the data interface of camera sensor or as the motor drive interface.
5	S2 switch	Function selection switch of GPIO65, GPIO66, GPIO67, and GPIO6 interfaces, which can be used as the data line of the camera sensor or as the motor drive interface.
6	CON4 sensor interface	Used to connect the sensor board.
7	JP2, JP3 jumper caps	Used to select the power supply for the MIPI interface or DVP interface.
8	JP4, JP10 jumper caps	Used to select USB Wi-Fi module or USB burning.
9	Power switch	The power switch of the development board.
10	USB interface	Standard Micro USB interface, used for USB power supply and program burning. Note: JP4 and JP10 need to be configured in USB burning mode.
11	Adapter (5V DC)	Power supply for the development board.
12	Wi-Fi Module	USB Wi-Fi module.
13	Core board interface	Used to connect the core board.
14	Network Interface	RJ45 interface for wired network.
15	Serial Port	The default print output serial port can be used for development board debugging.
16	PHY/Wi-Fi board interface	Interface for connecting to PHY or SDIO Wi-Fi board.
17	Flash	SPI NOR Flash of the development board.
18	TF card holder	Standard TF card slot, supports up to 128GB.
19	S5 switch	The function selection switch of GPIO41 and GPIO42 can be used as the CLK and CMD interface of the SD card, or as the audio enable SPK_EN and plug-in detection.
20	BOOT button	Used for burning development boards.
21	RESET button	Reset button.

Number	Name	Illustrate
22	RECOVER button	Soft reset button, retains the camera network configuration (DHCP, IP, subnet, gateway), and restores other configurations to factory settings.
23	Reserve 1	Reserved, no function is defined yet.
24	Reserve 2	Reserved, no function is defined yet.
25	MIC interface	Microphone input interface.
26	S6 Switch	The HPOUT interface's function selection switch lets you choose between speaker output and headphone jack output.
27	JTAG interface	JTAG debug interface.
28	Line-in interface	3.5mm Line-in input.
29	Headphone output interface	3.5mm mono headphone output.
30	Speaker interface	For connecting speakers.
31	CON9 sensor interface	Used to connect the sensor board.

3.2.1.2 Jumper cap description

There are jumper caps JP2, JP3, JP4, and JP10 on the bottom board of the development board. Please refer to Table 3-1 for the corresponding functions.

Note: The jumper caps are correctly configured at the factory. Please do not change them without authorization, otherwise the development board may not work properly or the components on the development board may be damaged.

3.2.1.3 Indicator Light Description

There are two indicator lights on the bottom plate of the development board, one for power supply and the other for working status indication.

Table 3-3 Indicator Light Description

Indicator light name	Logo/ Location	Function
Power indicator	D2	Development board power indicator (red): ● When the power is on, the red indicator light is on

Indicator light name	Logo/ Location	Function
		When the power is off, it goes off
Wi-Fi indicator System upgrade indicator	D7	Indicates the status of Wi-Fi network and system upgrade (blue) : The platform is not yet defined

3.2.2 Core board

The actual picture of the core board is shown in the figure below.

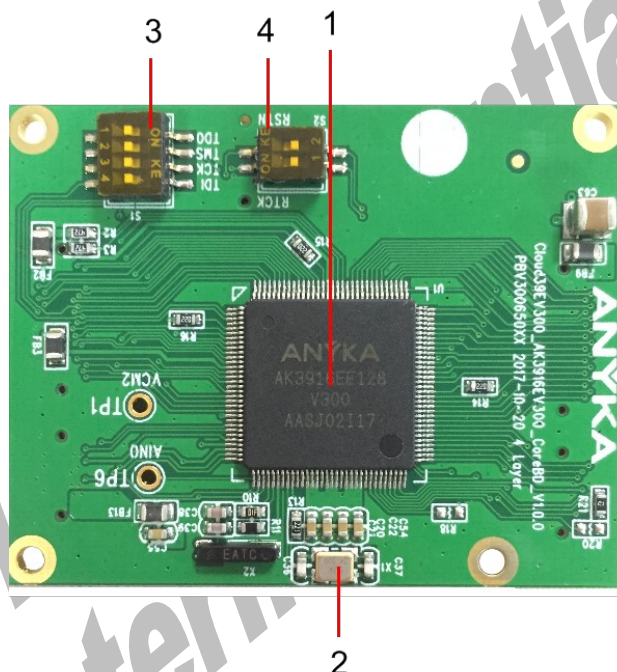


Figure 3-3 Core board physical picture

3.2.2.1 The core board interface and function description are shown in the following table.

Table 3-4 Core board interfaces and functions

Number	Name	Illustrate
1	Main control chip	The main control chip of the core board: AK3916EV300, AK3918EV300.
2	24MHz crystal	Used to provide the clock for the main control chip.
3	S1 switch	Function selection switch for GPIO64, GPIO70, GPIO71, and GPIO72 interfaces. Select to use for sensor interface or JTAG interface.

Number	Name	Illustrate
4	S2 switch	Function selection switch of GPIO5 and GPIO0 interfaces, select to use for JTAG interface or for other IO functions.

3.2.3 Expansion Board

The expansion boards that come with the development board include a sensor board, a PHY board, and an SDIO Wi-Fi board. The actual pictures are shown in the figure below.

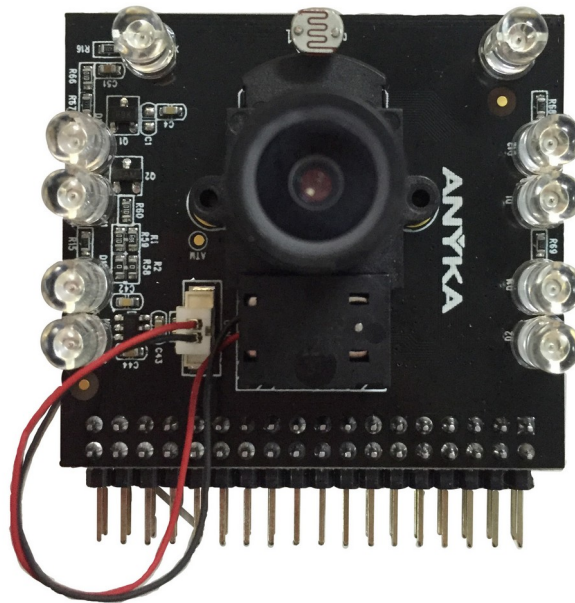


Figure 3-4 Sensor board

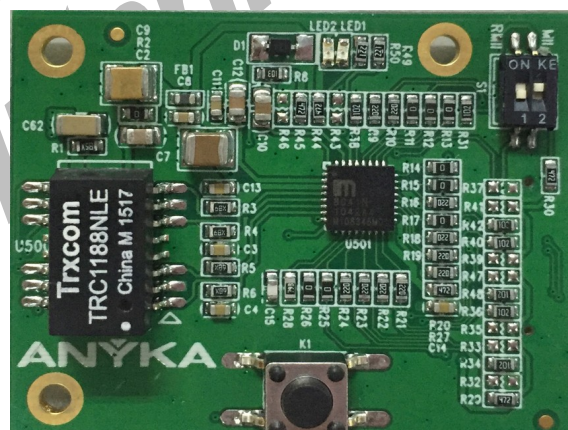


Figure 3-5 PHY board



Figure 3-6 Wi-Fi board

4 Development board debugging

The development board can be debugged through the serial port. The connection steps are as follows.

- (1) Use RS232 serial cable to connect the development board UART0 to the PC serial port;
- (2) Run the Xshell software, the software interface is as follows:

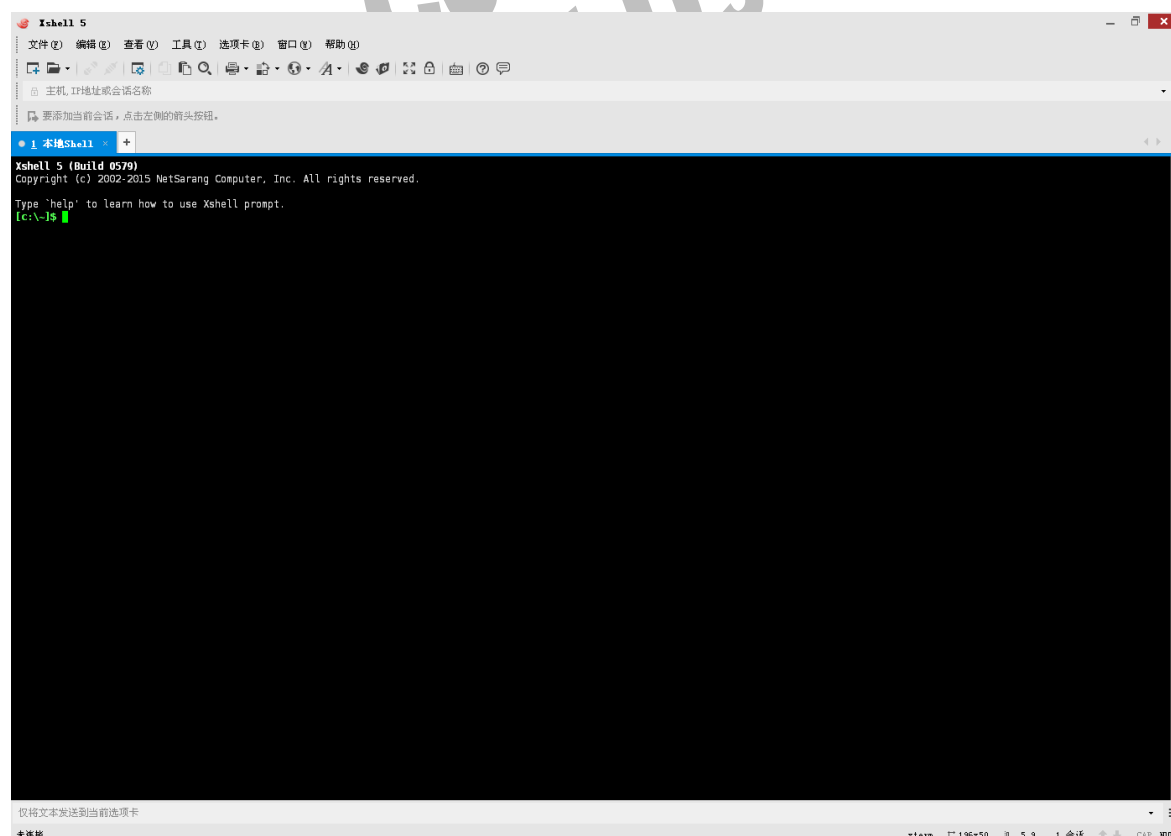


Figure 4-1 Xshell software interface

(3) Click the New button in the upper left corner. In the pop-up dialog box, select the "Connection" option on the left and select "SERIAL" in the Protocol column.



Figure 4-2 New session interface

(4) (4) Select the "SERIAL" option on the left, and select the corresponding serial port, bit width, and baud rate.

(5) Note: The default baud rate is 115200. The port can be viewed in the "Device Manager" on the PC.

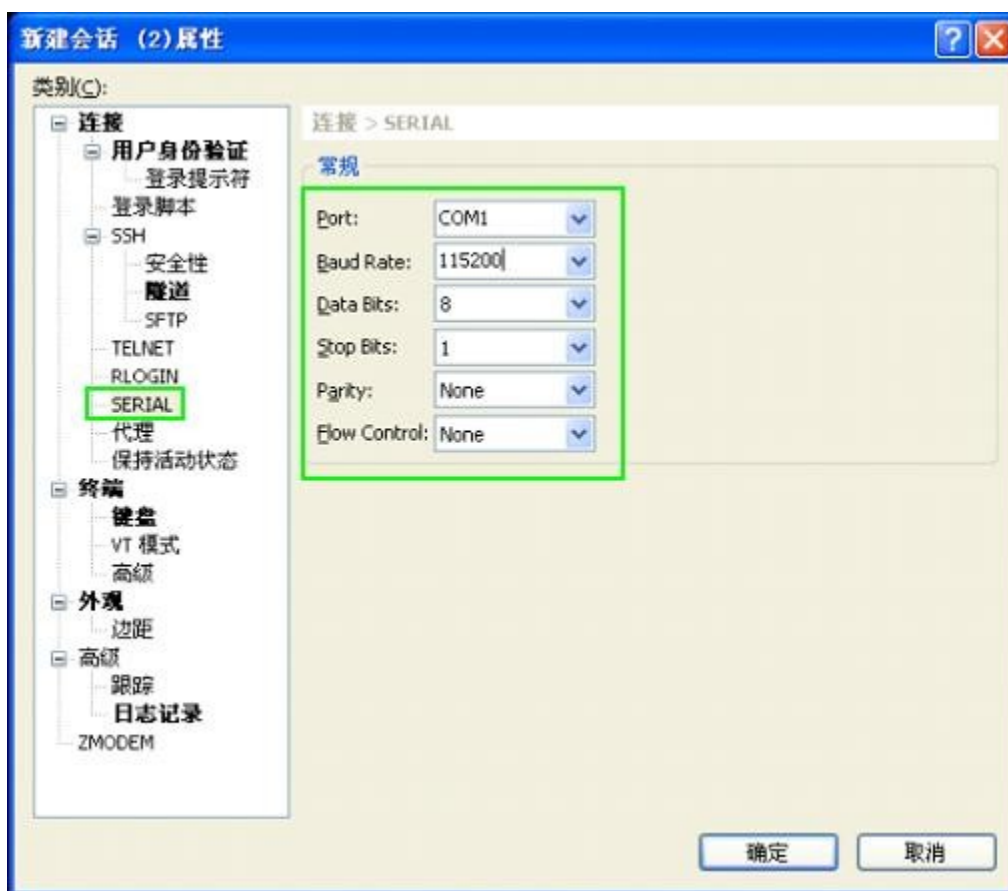


Figure 4-3 SERIAL parameter configuration interface

(6) Select Logging under Advanced to set the log saving path.

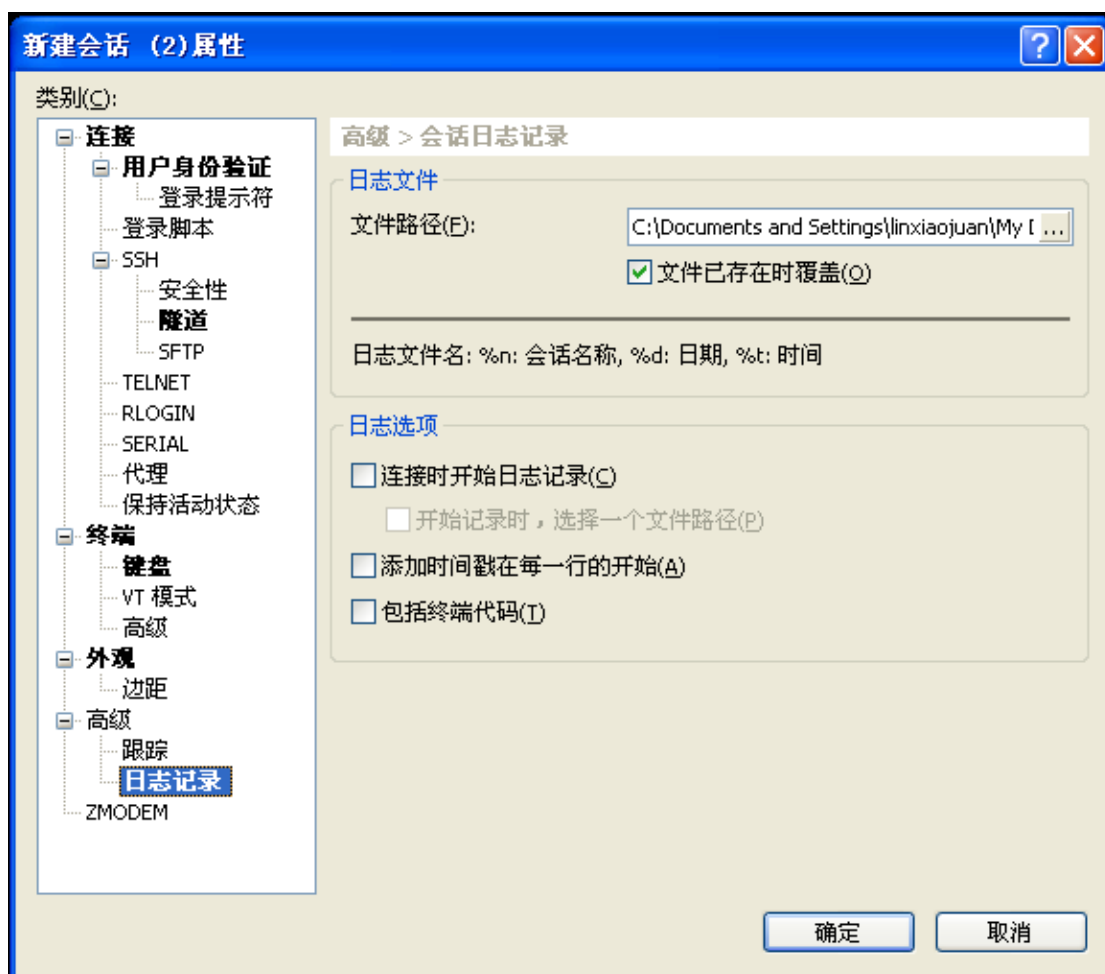


Figure 4-4 Logging interface

(7) Plug one end of the USB data cable into the USB port of the development board and the other end into the USB port of the PC. At this time, the UART0 serial port outputs the corresponding print information. The account to enter the platform debugging state is root, and the command is cloud39ev300.

(8) In order to enable users to better master platform development, the Cloud39EV300 platform provides a relatively comprehensive supporting Demo program to demonstrate the application examples of middleware functions. Users can refer to the interface calling method in the Demo to clarify the functions and usage of the interface, so as to facilitate secondary development according to their own needs. For detailed Demo operation instructions, please refer to the platform "Cloud39EV300 Platform User Development Manual".

5 Development board burning

The steps for burning the development board are as follows.

- (1) Make sure the JP4 and JP10 jumpers on the baseboard are configured for USB programming mode, namely: USB_DP->AK_DP, USB_DM->AK_DM.
- (2) Run the Burn Tool on the PC, make sure the Auto Burn option in the operation interface is checked, and the partition size and platform options in the Burn Tool are correctly configured.

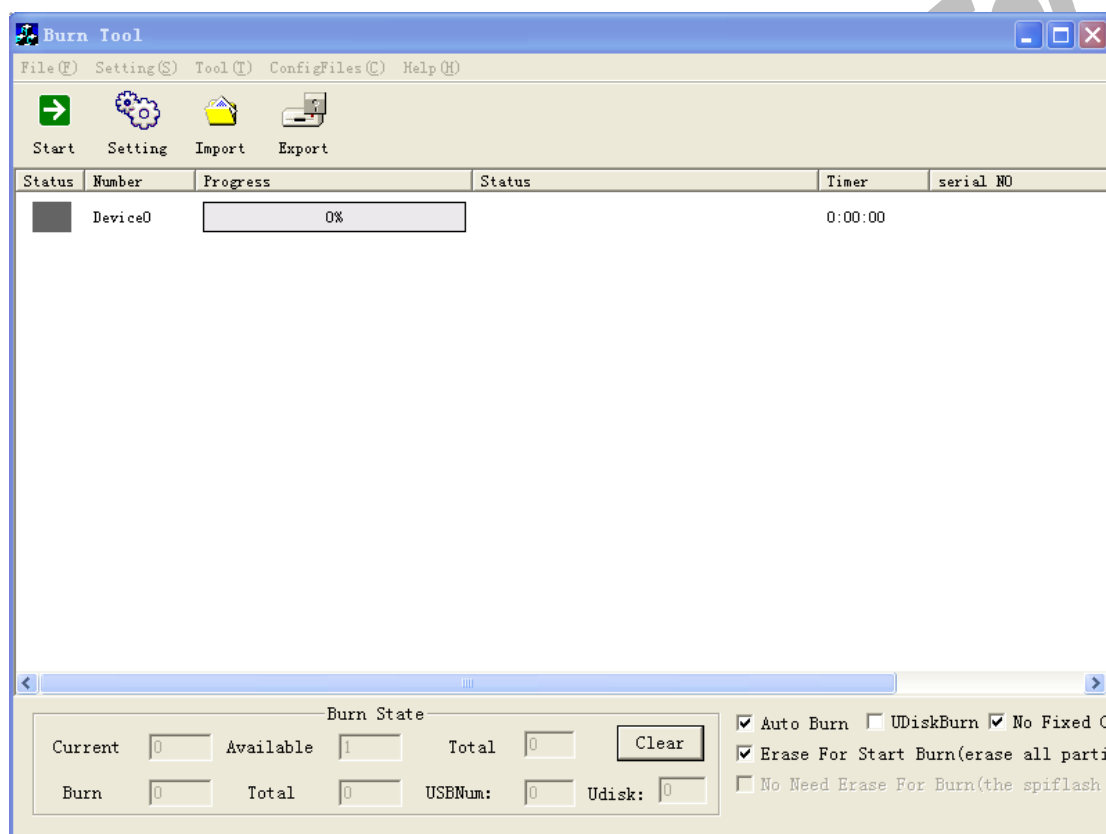


Figure 4-1 Burning tool operation interface

- (3) First, use a USB cable to connect the PC and the development board, hold down the BOOT button, and press the power button to power on the board (note that if the board is already powered on after the USB connection, that is, the power button is on, you can short press the reset button to achieve the same effect). The board will automatically burn, and you can release the BOOT button after the channel status of the burning tool turns yellow.

- (4) Wait for the progress bar in the burning tool to complete and the development board is burned.

(5) Note: For the use of the burning tool, please refer to the platform PDK supporting document "Burning Tool User Manual". For the method of generating the development board image program, please refer to the platform PDK supporting document "Cloud39EV300 Platform User Development Manual".

6 Network Configuration

The development board supports wired and wireless networking. The specific configuration method is as follows.

6.1 Wired network

The development board will automatically enable the wired network mode after connecting to the wired network. If the wireless Wi-Fi connection has been used before, the system will automatically turn off the Wi-Fi.

When the development board is in the serial port debugging state, use the `ifconfig` command to set the network parameters of the development board's network card, and enable or disable the network card. For the usage and instructions of `ifconfig`, you can enter `ifconfig -help` to view it. The following only takes setting the network card address as an example.

Assuming that the Ethernet network card address of the development board needs to be set to 192.168.1.100, execute the following command:

```
ifconfig eth0 192.168.1.100
mac init success!
eth0: link down
eth0: link up, full duplex, 100Mb
```

If the network card is turned off, it will be automatically turned on after the network address is set successfully.

If you want to disable the network card, execute:

```
ifconfig eth0 down
```

If you want to manually enable a disabled network card, execute:

```
ifconfig eth0 up
```

6.2 Wireless network

The development board can support Wi-Fi modules with SDIO interface and USB interface. A Realtek RTL8188 USB Wi-Fi module is installed on the baseboard by default.

Regarding the use of the Wi-Fi module, please note the following:

(1) The system platform configuration uses the USB interface Wi-Fi module by default. By default, the JP4 and JP10 jumper caps on the baseboard are connected to the USB Wi-Fi module WIFI_DP->AK_DP, WIFI_DM->AK_DM. Please make sure that the connection method of the baseboard jumper cap is correct (see 3.2.1.1.2 Jumper Cap Description section for details).

(2) If you need to use a Wi-Fi module with an SDIO interface on the development board, you need to do the following:

- Remove the PHY board and replace it with a Wi-Fi board. The platform currently does not support SDIO-WiFi modules.
- Modify the configuration file and select the SDIO Wi-Fi module.
- Note: For configuration information related to the Wi-Fi function, please refer to the Cloud39EV300 Platform Configuration Instructions.

(3) Manually set the SSID and password to allow the device to access the specified router.

The steps are as follows:

- Open the configuration file, vi /etc/jffs2/onvif_cfg.ini (ONVIF protocol) or anyka_cfg.ini (other platforms).
- Modify the ssid and password of the [wireless] option in the configuration file and enter the account and password of the corresponding router.
- Use the reboot command to restart the development board. After restarting, if there is no wired network connected, the development board will automatically connect to the corresponding router.

7 Video Preview

In addition to the VLC and Potplayer software mentioned in Chapter 2, the development board supports the standard ONVIF protocol and can use the mainstream NVR on the market (must support H.265 decoding) for video preview. It should be noted that the IP address of the NVR must be in the same network segment as the IP address of the development board, otherwise the preview will fail. To use the NVR for preview, please follow the NVR operation steps, which will not be repeated here.

In addition, the development board can also configure the encoder to H.264 mode, and then use ONVIF Device Manager to preview the video. The configuration and preview steps are as follows.

(1) View the configuration file in serial port debugging mode. The path is `vi /etc/jffs2/onvif_cfg.ini`. Make sure that the values of the parameters `main_enc_type` and `sub_enc_type` are both 1 (1 -> H.264, 2 -> H.265, 3 -> H264+, 4 -> H265+). If not, set it to 1 and save and exit the configuration file, and restart the development board.

Note: If you are not using the ONVIF protocol version, H.265+/H.264+ functions are not supported yet. To switch the encoder, please use the `anyka_cfg.ini` configuration file in the same path, and configure the parameters: 0->H.264, 1->MJPEG, 2->H.265.

(2) The default Ethernet IP of the device is 192.168.1.88. Make sure that the development board and the PC are in the same LAN. Regardless of whether the device uses a wired or wireless connection, please connect the device and the PC to the same router.

(3) Run the ONVIF Device Manager program. The ONVIF test tool automatically searches for video devices in the LAN, and the search results are displayed under the Device List, as shown in the figure below.

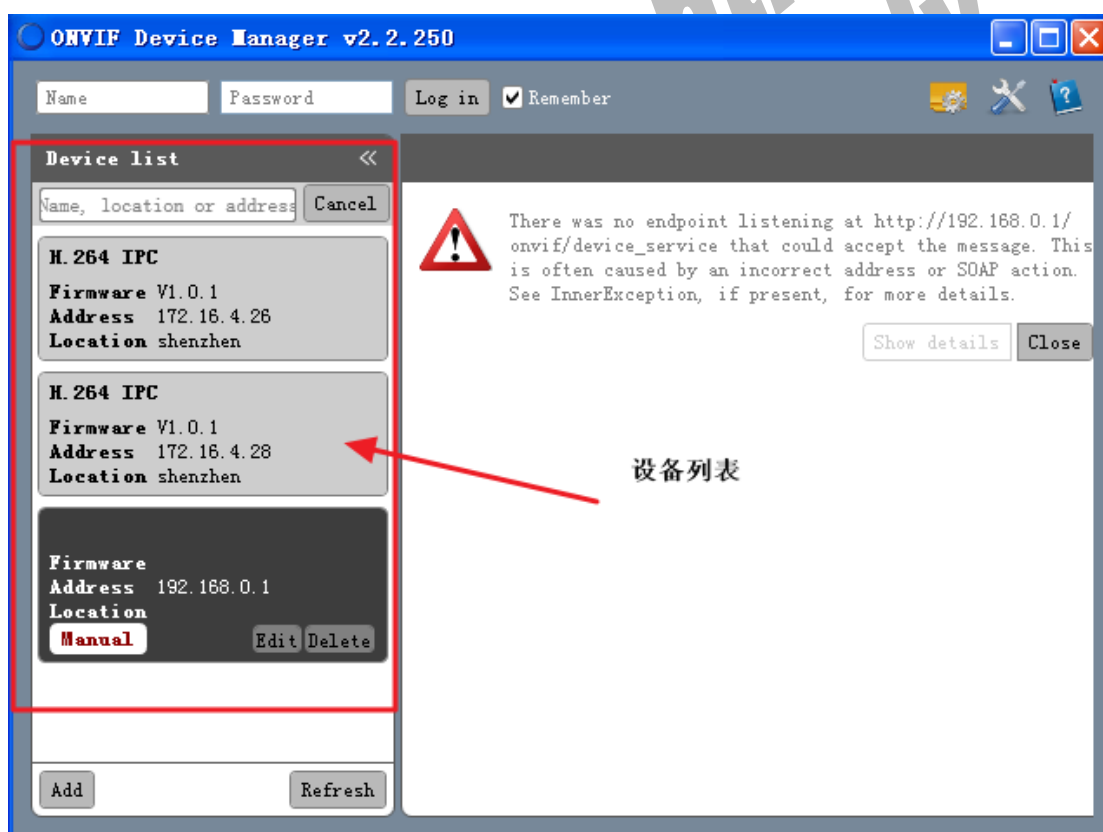


Figure 7-1 Main interface of ONVIF test tool

(1) If no device is found, you can click the Add button to add it manually. Click the Refresh button to refresh the device list. Click the specified device to display the relevant menu in the right view.

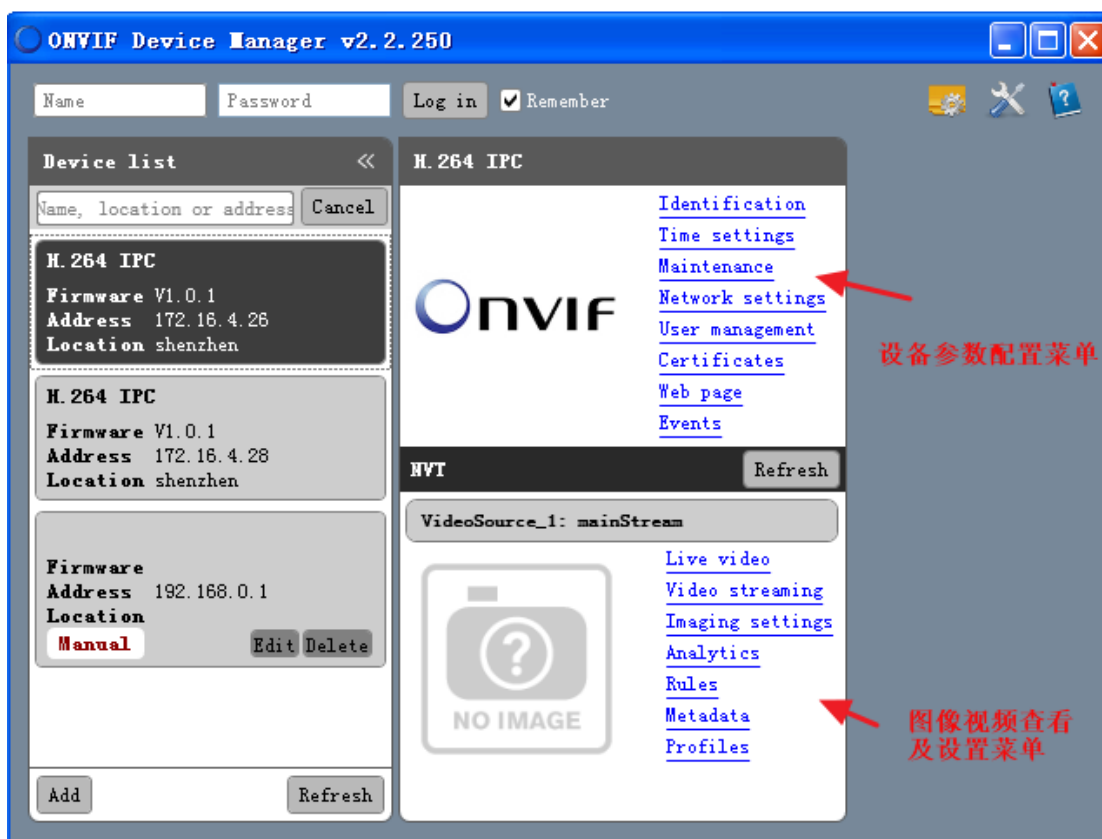


Figure 7-2 ONVIF test tool device related pages

(2) Click the Live Video button to view the live video.



Figure 7-3 Viewing real-time video using ONVIF test tool

8 Development Board Use Precautions

When operating the development board, it is recommended that users wear anti-static gloves or anti-static wrist straps to prevent the development board from being damaged by static electricity during use.

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